

Q1. DIRECTIONS *for question 1:* Type in your answer in the input box provided below the question.

Find the number of solutions to the equation $6x^2 + 5x - 6 - e^x = 0$.

Q2. DIRECTIONS *for questions 2 to 6:* Select the correct alternative from the given choices.

Ram and Sita had a son named Bala. The sum of the present ages of Ram and Bala is 120 years. When Ram was as old as Sita is, Sita was twice as old as Bala. Find the present age of Bala (in years).

☐ a) 35

☐ b) 30

☐ c) 40

☐ d) 20

Q3. DIRECTIONS *for questions 2 to 6:* Select the correct alternative from the given choices.

If $a = \sqrt{17} + \sqrt{8}$, $b = \sqrt{14} + \sqrt{11}$, $c = \sqrt{16} + \sqrt{9}$, and $d = \sqrt{13} + \sqrt{12}$, which of the following is true?

- ☐ a) $b > d > a > c$
- ☒ b) $d > b > c > a$ ✓ Your answer is correct
- ☐ c) $d > c > b > a$
- ☐ d) $a > c > b > d$

Q4. DIRECTIONS for questions 2 to 6: Select the correct alternative from the given choices.

In a triangle ABC, let AD, BE and CF be the medians to BC, CA and AB respectively and let G be the point of intersection of the medians. If the quadrilaterals AFGE, BFGD and CDGE are represented as Q_1 , Q_2 and Q_3 respectively, then which of the following statements is/are true?

- I. If Q_3 is concyclic, then $AE = \sqrt{3} GD$.
- II. If Q_1 , Q_2 and Q_3 are all concyclic, then triangle ABC is right-angled.
- III. If Q_1 , Q_2 and Q_3 are all concyclic, then triangle ABC is equilateral.

- ☐ a) I and II only
- ☐ b) I and III only
- ☐ c) Only III
- ☐ d) Only I

Q5. DIRECTIONS *for questions 2 to 6:* Select the correct alternative from the given choices.

By selling two items, each at the same selling price, a shopkeeper made an overall profit of 20%. If one item was sold at a profit of 50%, then the other was sold at

- ☐ a) a loss of 10%.
- ☐ b) a loss of 30%.
- ☐ c) a profit of 10%.
- ☒ d) no profit no loss. ✓ Your answer is correct

Q6. DIRECTIONS *for questions 2 to 6:* Select the correct alternative from the given choices.

Let m be a positive integer greater than 4 and $n = m^2 - m$. Then $n^3 - 6n^2 + 8n$ is always divisible by

- ☐ a) 48.
- ☐ b) 6.
- ☐ c) 12.
- ☐ d) 124.

Q7. DIRECTIONS *for question 7:* Type in your answer in the input box provided below the question.

Arif, Bill and Chetan were classmates and wanted to watch a movie at South City Mall right after their class got over. Bill had an Activa bike which could accommodate a maximum of two persons at a time and they wanted to reach the mall in the minimum time possible. They started simultaneously from their college, Bill and Arif on the bike and Chetan on foot. Bill dropped Arif at a certain point along the route to the mall, turned back and on his way met Chetan. He picked up Chetan and then traveled towards the mall such that all three of them reached the mall simultaneously. What is the total time (in minutes) taken by them to reach the mall from their college, if the speed of the bike was 30 km/hr, the walking speed of each of them was 6 km/hr and the distance between their college and the mall was 22.5 km?

Q8. DIRECTIONS *for question 8:* Select the correct alternative from the given choices.

P1 and P2 are two sets of parallel lines, such that the number of lines in P2 is greater than that in P1. If they intersect at a total of 18 points, the total number of parallelograms formed by P1 and P2 cannot be

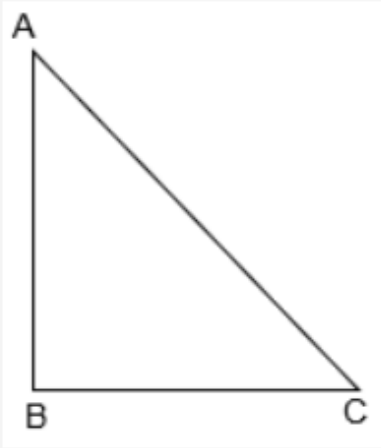
- ☐ a) 36.
- ☐ b) 45.
- ☐ c) 54.
- ☐ d) All the above

Q9. DIRECTIONS *for question 9:* Type in your answer in the input box provided below the question.

If the maximum value of $\sin^2\theta - 2\sin^3\theta$ is $\frac{1}{A}$, for $0^\circ < \theta < 90^\circ$, find the value of A .

Q10. DIRECTIONS for question 10: Select the correct alternative from the given choices.

In the right-angled triangle given below, $AB = BC = 6\sqrt{2}$ cm. The side AB is divided into 13 equal parts by drawing 12 line segments, each parallel to BC, from AB to AC. Find the sum of the lengths of these 12 line segments.



☐ a)
 $36\sqrt{2}$ cm

☐ b)
 $39\sqrt{2}$ cm

☐ c)
 $42\sqrt{2}$ cm

☐ d)
 $72\sqrt{2}$ cm

Q11. DIRECTIONS *for questions 11 to 14:* Type in your answer in the input box provided below the question.

If a and b are both natural numbers, and $\log_2 (a \log_3 b) = 1$, how many ordered pairs (a, b) exist?

Q12. DIRECTIONS *for questions 11 to 14:* Type in your answer in the input box provided below the question.

If $2^{100} - 2^{99} - 2^{98} \dots\dots\dots - 2^{50} = k \times 2^n$, where k and n are natural numbers, what is the greatest possible value of n ?

Q13. DIRECTIONS *for questions 11 to 14:* Type in your answer in the input box provided below the question.

The sum of the first 12 terms of a geometric progression is six times the sum of the next six terms. If the first term of the progression is 32, what is the seventh term of the progression?

Q14. DIRECTIONS *for questions 11 to 14:* Type in your answer in the input box provided below the question.

The units digit in the expansion of $[(999)^{999}]^{999}$ is

Q15. DIRECTIONS *for questions 15 and 16:* Select the correct alternative from the given choices.

Two vessels, A and B, contain spirit and water, mixed in the ratio 7 : 3 and 4 : 9 respectively. Find the ratio in which the contents of A and B should be mixed so as to obtain a new mixture, containing spirit and water in the ratio 5 : 8.

- ☐ a) 12 : 49
- ☒ b) 10 : 41 ✓ **Your answer is correct**
- ☐ c) 11 : 45
- ☐ d) 9 : 37

Q16. DIRECTIONS *for questions 15 and 16:* Select the correct alternative from the given choices.

Nitish was to get a 50% hike in his pay but the computer operator wrongly keyed in the figure as 80% and printed the new pay slip. He received salary at this level for three months before the company realized the mistake. What percentage of his correct new salary will he get in the fourth month, if the excess salary paid to him in the previous three months is deducted in the fourth month?

- ☐ a) 20%
- ☐ b) 25%
- ☒ c) 40% ✓ **Your answer is correct**
- ☐ d) $33\frac{1}{3}\%$

Q17. DIRECTIONS *for question 17:* Type in your answer in the input box provided below the question.

Ajay had N coins with him and he gave one more than half the number of coins with him to Bulu and then gave two more than one-third of the remaining number of coins to Chintu. If N is a two-digit number, how many distinct values can N assume?

Q18. DIRECTIONS *for question 18 to 22:* Select the correct alternative from the given choices.

If the sum of the units digits in P^Q and 5^P is Q , where P and Q are natural numbers, such that $P < 4$ and $Q < 7$, how many possible values exist for Q^P ?

☐ a) 3

☐ b) 2

☐ c) 1

☐ d) 0

Q19. DIRECTIONS *for question 18 to 22:* Select the correct alternative from the given choices.

In rectangle ABCD, a semicircle is constructed with AB as the diameter. The semicircle intersects CD at points P and Q. M is a point on AB such that PM is perpendicular to AB. If the perimeters of rectangles AMPD and MBCP are 18 cm and 36 cm respectively, find the area (in sq.cm) of the semicircle.

- ☐ a) 28.125π
- ☐ b) 32.625π
- ☐ c) 36.375π
- ☐ d) 38.125π

Q20. DIRECTIONS *for question 18 to 22:* Select the correct alternative from the given choices.

If $f(x) = f(x - 1) - f(x + 1)$, while $f(1) = 3$ and $f(4) = 1$, what is the value of $f(7)$?

☐ a) -1

☐ b) -2

☐ c) 0

☐ d) 2

Q21. DIRECTIONS *for question 18 to 22:* Select the correct alternative from the given choices.

Find the quadratic equation whose roots are half the roots of the equation $x^2 + 5x + 3 = 0$.

☐ a) $x^2 + 10x + 12 = 0$

☐ b) $4x^2 + 20x + 3 = 0$

☐ c) $x^2 + 9x + 7 = 0$

☐ d) $4x^2 + 10x + 3 = 0$

Q22. DIRECTIONS *for question 18 to 22:* Select the correct alternative from the given choices.

How many positive integral values of n exist, such that $\frac{n + 76}{n + 4}$ is an integer?

☐ a) 5

☐ b) 6

☐ c) 7

☐ d) 8